Static Gesture Classification — Run Report

Generated 2026-05-16 14:03:56

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 31 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 30 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 30 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 30 trials
Total training paths	10 · 295 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports/RandomForest

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	22 channels
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✗	✗	✓	✗
left index	✗	✗	✓	✗
left middle	✗	✗	✓	✗
left ring	✗	✗	✓	✗
left pinky	✗	✗	✓	✗
left wrist	✗	—	—	—
right palm	✓	—	—	—
right thumb	✗	✗	✓	✗
right index	✗	✗	✓	✗
right middle	✗	✗	✓	✗
right ring	✗	✗	✓	✗
right pinky	✗	✗	✓	✗
right wrist	✗	—	—	—

Enabled: 12 · Disabled: 32 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 1188
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=1180 · test=59

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M- = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.*

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Prec	W-Rec	W-F1
Random Forest	1.0000	0.0000	1.0000	0.0000	0.9831	0.9848	0.9815	0.9823	0.9846	0.9831	0.9830

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_Lower	0.91	1.00	0.95	10
Double_Pistol_Recoil	1.00	1.00	1.00	10
Double_Raise	1.00	1.00	1.00	10
Double_Wiggle	1.00	1.00	1.00	10
Double_cmere	1.00	1.00	1.00	10
Drum_Roll	1.00	0.89	0.94	9
accuracy			0.98	59
macro avg	0.98	0.98	0.98	59
weighted avg	**0.98**	**0.98**	**0.98**	**59**

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000

Per-class external accuracy

Class	Random Forest	Support
Double_Lower	1.000	5
Double_Pistol_Recoil	1.000	5
Double_Raise	1.000	5
Double_Wiggle	1.000	5
Double_cmere	1.000	5
Drum_Roll	1.000	5